
Group Assignment

Level: 05

Cloud Infrastructure

Batch Code: COM2521
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Executive Summary

MedSys Health Solutions Ltd supports more than 40 private clinics through a web portal that holds appointments, patient records, diagnostic reports and medical images. Its infrastructure has grown piece by piece into a mix of legacy systems and virtual servers, run by a small internal team and external contractors. The result is a platform that slows at busy times, depends on manual backups, and would be hard to recover after a ransomware attack like the one that disrupted NHS pathology services in London in 2024 (NHS England, 2024).

This report recommends a phased move to Amazon Web Services (AWS) in the London Region, with **Amazon S3** as the foundation for every patient document and image. The portal runs on Amazon EC2 across two Availability Zones behind a load balancer, structured data moves to Amazon RDS, and email moves to Amazon SES. S3 versioning, Object Lock, cross-Region replication and AWS Backup replace the manual backup routine and keep copies that ransomware cannot change. Migration runs in five phases over about six months, with data synchronised continuously so that the final switch-over needs only a short overnight window.

Section 8 sets out five recommendations. The most urgent are to confirm the assumptions in Appendix A with the IT manager, complete a Data Protection Impact Assessment, and move files to S3 before the portal itself.

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1. Introduction

MedSys Health Solutions Ltd (MedSys) is a private healthcare services provider founded in 2008 in the West Midlands. It gives a growing network of private clinics and independent practitioners digital services for appointment scheduling, patient record management and the secure handling of diagnostic reports. Management has asked whether its on-premises, virtualised infrastructure can keep supporting growth, reliability and security, and whether cloud computing would be a better foundation.

Cloud computing is used here in the NIST sense: on-demand, self-service access to a shared pool of computing resources that can be provisioned and released quickly with minimal management effort (Mell and Grance, 2011). Acting as an independent cloud infrastructure consultant, this report:

- analyses the current environment and the problems it causes (Section 2);
- evaluates the benefits, risks and compliance implications of a cloud strategy (Section 3);
- researches **Amazon Simple Storage Service (Amazon S3)**, the technology specified for this engagement (Section 4);
- recommends and justifies an AWS architecture and topology (Section 5);
- sets out a low-downtime migration plan and the changes to IT operations (Section 6), followed by benefits and risks (Section 7).

1.1 Clarification Meeting and Assumptions

The brief allows one 20-minute meeting with the IT manager. Appendix A lists the nine questions prepared for that meeting (Table A1), chosen because their answers most change the design, and the eight assumptions this report relies on until they are confirmed (Table A2). The assumptions with the greatest effect on the design are:

- about 15 TB of files today, growing 20–25% a year (A1);
- the portal can run on Linux virtual machines without major code changes (A2);
- a recovery time of 4 hours and data loss of no more than 15 minutes, with no loss of uploaded documents (A4);
- primary data kept in the UK, with a disaster-recovery copy in Ireland acceptable subject to a DPIA (A6).

2. Current Infrastructure Analysis

2.1 Technical Analysis

MedSys runs a mixture of legacy systems, virtualised servers and internally hosted services added as needs arose rather than to one design. Three technical problems follow from this.

Capacity is fixed. Virtual machines can only use the physical hosts beneath them, so the portal is sized for an average day and slows when clinic activity peaks during business hours. Adding capacity means buying and installing hardware, which takes weeks rather than minutes, and that capacity sits idle overnight.

File storage grows without a plan. Clinics upload documents, reports and medical images every day, and these must be kept for years. On-premises storage has to be bought ahead of demand, and a file uploaded today sits on the same expensive disks as one uploaded seven years ago that nobody has opened since.

There are single points of failure. One server room, storage array or internet link can take the portal offline for every clinic at once. Legacy components are also harder to patch and integrate, which slows new features that growing clinics expect.

2.2 Operational Analysis

IT work is split between a small internal team and external contractors. Without shared documentation and standard builds, systems drift apart and incident response depends on who happens to be available. Routine tasks such as patching, capacity planning and backups absorb time that could be spent improving services.

Maintenance is done outside working hours to avoid disrupting clinics, which means overtime and rushed change windows. Backups are copied to an external location that MedSys manages itself. As data grows, each backup takes longer, and the brief gives no evidence that restores are tested; a backup that has never been restored is an unknown risk.

Email is business-critical for clinic coordination and patient notifications, yet mailing lists and contact details are held alongside the core systems. A failure of those systems also stops notifications, and personal data is spread across more places than necessary.

2.3 Security Analysis

Health data is **special category data** under UK GDPR and the Data Protection Act 2018, and it needs additional safeguards (ICO, n.d.). Management is right to worry about external threats: the June 2024 ransomware attack on Synnovis, a pathology supplier, disrupted blood tests and planned procedures across several London hospitals for weeks (NHS England, 2024). The current set-up is exposed in five ways:

- backups reachable from the main network can be encrypted or deleted by the same attack as the live data;
- legacy systems are harder to patch consistently, which widens the attack surface;
- access is shared between staff and contractors, with limited central logging of who did what;
- email carrying patient information is a common route for phishing and data leaks if it is not protected with SPF, DKIM, DMARC and encryption in transit;
- nothing automatically detects unusual activity or personal data stored in the wrong place.

2.4 Impact on Performance, Reliability and Growth

Table 1: How current issues affect the business

Issue	Performance	Reliability	Future growth
Fixed on-premises capacity	Portal slows during business-hour peaks	Overload can cause outages	Each new clinic needs new hardware

Issue	Performance	Reliability	Future growth
Unplanned storage growth	Large uploads compete with portal traffic	Full storage means failed uploads	Capital spend rises every year
Manual, self-managed backups	Backup jobs clash with evening work	Restore time unknown; copies exposed to ransomware	More data means longer backup windows
Mixed legacy estate, split team	Hard to tune end to end	Single points of failure; knowledge silos	Slow to add features and integrations
Security exposure	Incidents consume staff time	A breach could stop the service	Loss of clinic trust; regulatory action

Taken together, the current environment can keep today's service running but is poorly placed to absorb more clinics, larger files and a more hostile threat landscape.

3. Cloud Strategy Evaluation

3.1 Benefits of a Cloud Strategy

- **Elasticity:** capacity follows demand, so servers are added during business hours and removed overnight (Erl, Mahmood and Puttini, 2013).
- **Pay for use:** spending shifts from buying hardware years ahead to paying monthly for what is used, including storage priced per gigabyte.
- **Resilience by design:** each AWS Region has several isolated Availability Zones (AZs), so a service can survive the loss of a whole data centre.
- **Less routine work:** managed services take over hardware replacement, database patching and backups, freeing the small team for higher-value work.
- **Security tooling at scale:** encryption with customer-controlled keys, central identity, continuous threat detection and full audit logs, which would be costly for MedSys to build alone.
- **Faster change:** infrastructure as code creates identical test and production environments in minutes.

3.2 Risks of a Cloud Strategy

The main risks are misconfiguration (for example, a storage bucket made public by mistake), compliance and data residency, costs growing without governance, a skills gap in a small team, disruption during migration, reliance on internet connectivity, and dependence on one provider. Each is assessed with a mitigation in the risk register in Section 7.2.

3.3 Shared Responsibility

AWS is responsible for security *of* the cloud: facilities, hardware, networking and the software behind managed services. MedSys stays responsible for security *in* the cloud: classifying data, managing identities and access, choosing encryption settings, configuring networks and securing its own

application (Amazon Web Services, n.d.-f). Moving to AWS reduces MedSys's workload but does not transfer its accountability as data controller.

3.4 Components Suitable for the Cloud

Table 2: Suitability of MedSys components for cloud deployment

Component	Data sensitivity	Availability need	Suitability and approach
Patient documents, reports, medical images	High (special category)	High; must never be lost	Very suitable: Amazon S3 with encryption, versioning, Object Lock and replication
Clinic web portal	High	High in business hours	Suitable: EC2 Auto Scaling behind a load balancer; modernise later
Relational database	High	High	Suitable: Amazon RDS Multi-AZ
Email and mailing lists	Medium (personal data)	Medium to high	Suitable: Amazon SES; contact data kept in RDS
Backups and archive	High	Must be restorable	Very suitable: AWS Backup and S3 Glacier storage classes
Unused legacy systems	Varies	Low	Retire after exporting any data that must be kept

3.5 Compliance Considerations

- **UK GDPR and the Data Protection Act 2018:** MedSys needs a lawful basis and an Article 9 condition for health data, a Data Protection Impact Assessment (DPIA) before migration, and a data processing agreement with AWS.
- **Data residency:** primary data stays in London (eu-west-2). The disaster-recovery copy in Ireland must be recorded in the DPIA (A6); if contracts demand UK-only storage, recovery can rely on the London AZs plus locked backups instead.
- **Retention:** S3 Lifecycle and Object Lock enforce retention periods automatically and prove records have not been altered (A5).
- **Provider assurance:** the NCSC Cloud Security Principles give a checklist for assessing AWS (NCSC, n.d.). If MedSys takes on NHS-funded work, the Data Security and Protection Toolkit will also apply.

4. Specified Technology Research: Amazon S3

4.1 How Amazon S3 Works

Amazon S3 is AWS's object storage service. A file server organises data in folders on disks attached to a server. S3 instead stores each file as an **object**, made up of the data itself, a unique **key** (its name, such as clinics/0042/reports/2026/09/scan-8812.pdf) and **metadata** such as content type or clinic ID. Objects are held in **buckets**, and each bucket lives in one AWS Region chosen by the customer, so data does not leave that Region unless replication is switched on.

Applications read and write objects through a web API over HTTPS, either directly or through AWS software development kits. There are no disks to mount and no capacity to reserve: storage grows and shrinks with the data, and the customer pays per gigabyte-month stored plus requests and data transfer.

Behind the API, S3 Standard stores data redundantly across at least three AZs. AWS designs S3 for 99.999999999% ("eleven nines") durability and S3 Standard for 99.99% availability. S3 gives strong read-after-write consistency, so a report can be read the moment its upload finishes. Single objects can be up to 50 TB, and files over 100 MB should use multipart upload, which sends parts in parallel and retries only failed parts (Amazon Web Services, 2026a).

4.2 Key Functionality

Table 3: Amazon S3 features and their relevance to MedSys

Feature	What it does	Relevance to MedSys
Storage classes and Lifecycle	Moves objects between cost tiers by age or access pattern	New reports stay fast; old images cost less
Versioning	Keeps every version of an object; deletions can be undone	Recovery from accidental overwrite or deletion
Object Lock	Write-once-read-many: blocks deletion or overwrite until a retention date, in governance or compliance mode (Amazon Web Services, 2026d)	Tamper-proof records and ransomware-resistant copies
Encryption	All new objects encrypted by default (SSE-S3) since January 2023; SSE-KMS adds customer-managed keys and key-use logs (Amazon Web Services, 2026a)	Protects special category data at rest
Access control	IAM and bucket policies, Access Points, and Block Public Access, which is on by default for new buckets	Least-privilege access per application and clinic
Pre-signed URLs	Time-limited link that lets a browser upload or download one object without AWS credentials (Amazon Web Services, 2026e)	Clinics send large scans straight to S3
Event notifications	Emits an event when objects are created or deleted, which can trigger Lambda or EventBridge	Automatic malware scanning, indexing and alerts
Replication	Copies objects within or across Regions; Replication Time Control replicates 99.9% of objects within 15 minutes (Amazon Web Services, 2026b)	Disaster-recovery copy in a second Region
Visibility	CloudTrail data events, server access logs, S3 Storage Lens, Amazon Macie	Audit trail and discovery of misplaced personal data

Table 4: S3 storage classes and suggested use at MedSys

Storage class	Access characteristics	Suggested MedSys use
S3 Standard	Millisecond access, frequent use	Reports and images from the last 90 days
S3 Intelligent-Tiering	Moves objects between tiers automatically by access	Files with unpredictable access, such as attachments
S3 Standard-IA	Millisecond access, retrieval fee, infrequent use	Files 90 days to 1 year old
S3 Glacier Instant Retrieval	Millisecond access, rarely used data	Files 1–3 years old that must still open instantly if a patient returns
S3 Glacier Flexible Retrieval	Minutes to hours	Older backup sets
S3 Glacier Deep Archive	Within 12 hours; lowest cost	Records kept only to meet legal retention

4.3 How Organisations Use Amazon S3

S3 underpins data lakes for analytics, backup and archive, content delivery and application storage across many industries. Healthcare examples are the most relevant to MedSys:

- **GE Healthcare** built the GE Health Cloud on AWS so clinicians can share imaging across organisations. About one petabyte of imaging data is held in S3, which the company calls "the cornerstone of our solution" for durability and reliability (Amazon Web Services, n.d.-a).
- **King Hamad University Hospital and Bahrain Oncology Center** moved about one million imaging studies (48 TB) to S3. An S3 File Gateway connected their existing imaging system without re-engineering it; files land in S3 Standard and lifecycle rules move them to S3 Glacier Instant Retrieval, keeping millisecond access for clinicians and cutting archive storage costs by 33% (Dittforth and Allawi, 2022).
- **Genomics England** manages around 50 petabytes of genomic data from the 100,000 Genomes Project on AWS, relying on S3's durability and letting approved researchers analyse data without copying it (Amazon Web Services, n.d.-b).
- **Royal Orthopaedic Hospital NHS Foundation Trust**, in MedSys's own region of Birmingham, was the first in Europe to run GE's Edison True PACS imaging system in the cloud on AWS (NHS England Digital, n.d.).

The common pattern is S3 as a durable, low-maintenance system of record, a gateway that connects legacy systems during the transition, lifecycle rules that control cost, and new services built on the same data.

4.4 Benefits for MedSys Now

- **No more storage capacity planning:** S3 grows automatically as clinics join, and MedSys pays only for what it stores.
- **A faster, lighter portal:** pre-signed URLs let clinics upload large images directly to S3, so web servers no longer carry file traffic at peak times.

- **Protection against ransomware and error:** versioning, Object Lock in compliance mode and cross-Region replication replace the manual external backup. Locked copies cannot be deleted or encrypted even with stolen administrator credentials.
- **Lower cost for old data:** lifecycle rules move ageing images to Glacier storage classes automatically, the approach that saved King Hamad University Hospital 33%.
- **Compliance evidence:** KMS encryption, CloudTrail data events and Amazon Macie show who accessed which record and flag personal data stored in the wrong place.
- **Low-disruption migration:** S3 File Gateway lets legacy applications keep writing to a normal file share while the data lands in S3.

4.5 How S3 Could Help MedSys in Future

- **Analytics:** with S3 as a data lake, Amazon Athena and Amazon QuickSight can report on clinic activity and turnaround times without adding load to the live database.
- **Medical imaging:** AWS HealthImaging, available in the London Region since March 2026, stores and serves images at petabyte scale through standard DICOMweb interfaces (Amazon Web Services, 2026c). It is a natural next step if MedSys adds radiology viewing or image sharing.
- **Artificial intelligence:** Amazon Textract can extract data from scanned referral letters, and Amazon SageMaker models could predict missed appointments, both using data already governed in S3.
- **Partner and patient access:** S3 Access Points can give each new clinic or partner laboratory its own controlled entry point, and a future patient app can download reports using the same pre-signed URL pattern.

5. Proposed AWS Solution and Topology

5.1 Design Principles

The design follows the six pillars of the AWS Well-Architected Framework: operational excellence, security, reliability, performance efficiency, cost optimisation and sustainability (Amazon Web Services, n.d.-c). In practice this means:

- keep patient data in the UK, encrypt everything, and grant least-privilege access;
- remove single points of failure by running every tier across two AZs;
- prefer managed services so the small team runs less infrastructure;
- separate file storage (S3) from compute so each scales independently;
- build everything as code in separate production, test and security accounts governed by AWS Control Tower.

5.2 Recommended AWS Services

Table 5: Recommended AWS services and where they meet MedSys's needs

Layer	AWS services	Where it meets MedSys's needs
Governance	AWS Organizations, AWS Control Tower	Separate accounts; guardrails block public buckets and unapproved Regions
DNS and edge	Route 53, CloudFront, AWS WAF, AWS Shield	Secure, fast entry point; blocks common web attacks and DDoS; health-check failover
Network	VPC over two AZs, private subnets, NAT gateways, S3 gateway endpoint	Servers and database never exposed to the internet; S3 traffic stays on the AWS network
Compute	Amazon EC2 Auto Scaling, Application Load Balancer	Adds servers at busy times and removes them overnight; replaces failed servers automatically
Database	Amazon RDS for PostgreSQL, Multi-AZ	Automatic failover, patching and point-in-time recovery
File storage (specified technology)	Amazon S3 with Glacier storage classes	Durable store for documents and images with lifecycle, Object Lock and replication
Event processing	S3 Event Notifications, Amazon EventBridge, AWS Lambda	Malware scanning, metadata extraction and thumbnails without servers
Email	Amazon SES	Scalable notification email with SPF, DKIM and DMARC; bounce handling
Backup and DR	AWS Backup, S3 Cross-Region Replication, RDS cross-Region read replica	Automated, locked backups and pilot-light recovery in Ireland
Identity and keys	IAM Identity Center with MFA, AWS KMS, AWS Secrets Manager	Named access for staff and contractors; customer-managed encryption keys
Detection and compliance	GuardDuty, Amazon Macie, Security Hub, AWS Config, CloudTrail	Threat detection, personal data discovery, posture checks and a full audit trail
Monitoring	Amazon CloudWatch	Metrics, logs, alarms and scaling triggers
Migration (temporary)	Site-to-Site VPN, AWS DataSync, S3 File Gateway, AWS DMS	Secure transfer of files and database with continuous synchronisation

5.3 Proposed Topology

Figure 1 (next page) shows the target topology; the numbers below match its numbered flows.

1. Clinic staff connect over HTTPS through Route 53 and CloudFront, where AWS WAF filters malicious requests. The load balancer passes requests to portal servers in private subnets in two AZs; Auto Scaling keeps one server per zone and adds more as load rises.
2. The portal stores structured data in RDS PostgreSQL. A synchronous standby in the second AZ takes over automatically if the primary fails.
3. Files never pass through the web servers: after checking permissions, the portal returns a pre-signed URL valid for a few minutes, and the browser uploads or downloads directly with the bucket.

4. Servers reach S3 privately through a gateway endpoint; the bucket policy only allows the portal's role (which signs the URLs) and the processing functions.
5. Each new object triggers a Lambda function that scans it for malware, extracts metadata and writes an index record to RDS.
6. Lifecycle rules move ageing objects to cheaper Glacier storage classes.
7. S3 Cross-Region Replication and an RDS cross-Region read replica keep up-to-date copies in Ireland.
8. AWS Backup takes scheduled, vault-locked backups of RDS and S3 and copies them to the DR Region.
9. The portal sends clinic and patient emails through Amazon SES.
10. During migration only, files and database changes flow from the on-premises estate over a Site-to-Site VPN.

MedSys Health Solutions Ltd: Proposed AWS Architecture

S3-centred, multi-AZ design in the London Region with a pilot-light DR copy in Ireland | Prepared by M N M Shahmi (CB016484)

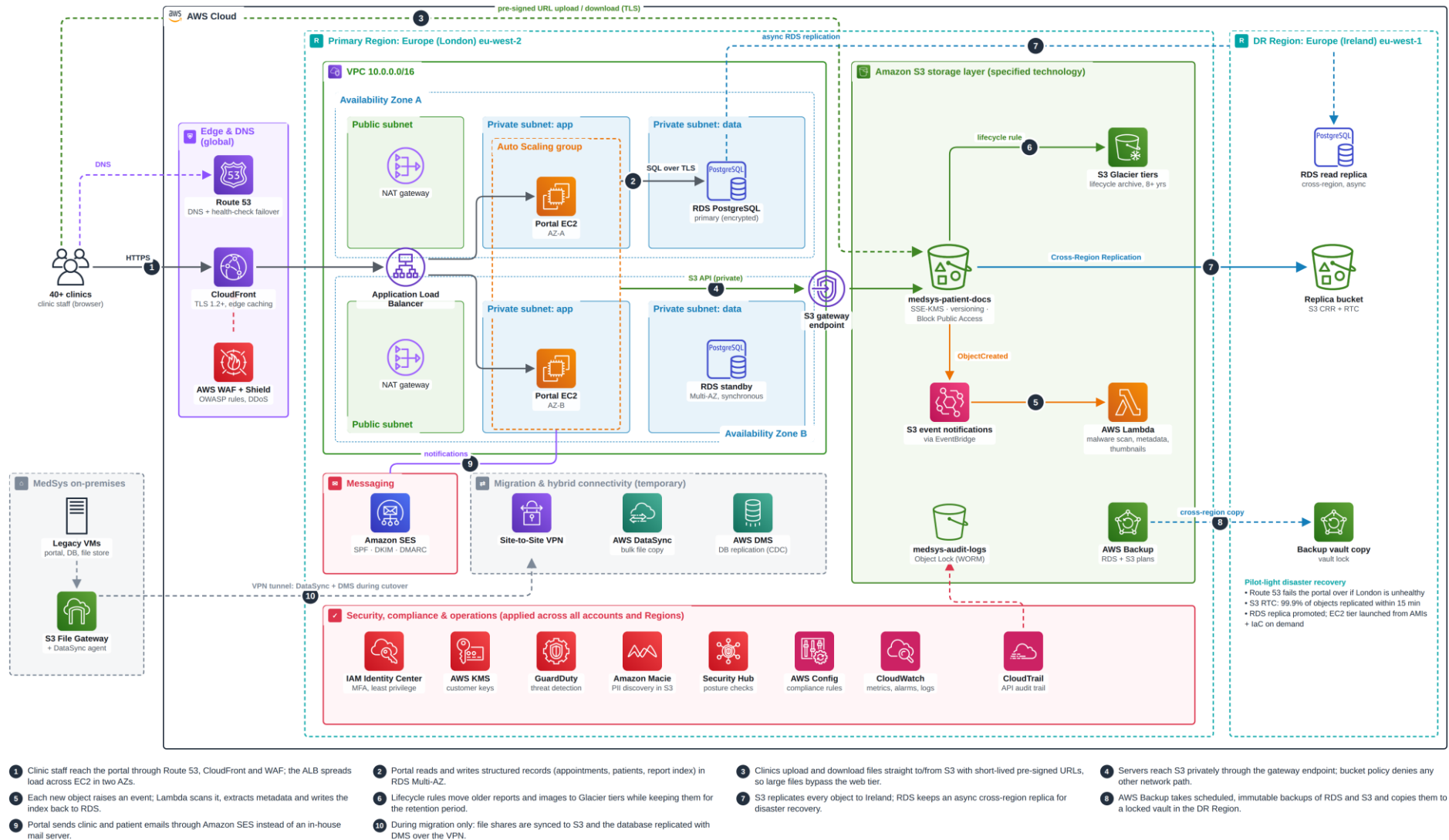


Figure 1: Proposed AWS architecture for MedSys (London primary Region, Ireland pilot-light DR Region)

5.4 Design Justification and Alternatives

Table 6: Key design decisions

Decision	Chosen option	Alternatives considered	Reason
Portal hosting	EC2 Auto Scaling (replatform)	Containers on ECS Fargate; fully serverless	Least code change for the first move (A2); containers can follow once on AWS
Database	RDS Multi-AZ	Database on EC2; Amazon Aurora	Managed patching and failover; Aurora can be adopted later if load justifies it
File storage	Amazon S3	Amazon EFS or FSx file shares	Cheaper at scale, lifecycle tiers, Object Lock, event triggers and eleven-nines durability
Email	Amazon SES	Keep on-premises mail server; third-party provider	Removes a server to patch, pay per message, stays under the same governance
Disaster recovery	Pilot light in Ireland	Backup and restore; warm standby; multi-site	Near-zero data loss with RTO of minutes to hours at modest cost (Amazon Web Services, n.d.-d), meeting A4
Primary Region	London	Ireland	UK data residency and lowest latency to West Midlands clinics

A pilot-light design keeps data continuously replicated to Ireland while compute there stays switched off. In a regional failure the RDS replica is promoted, the portal servers are launched from saved images using the same infrastructure code, and Route 53 moves traffic across. This costs much less than a warm standby while still protecting every uploaded document.

5.5 Cost Control

Costs are kept in check with Compute Savings Plans for the always-on baseline, Auto Scaling to remove unused capacity overnight, S3 Lifecycle and Intelligent-Tiering so older images do not sit on the most expensive tier, AWS Budgets alerts with cost-allocation tags, and a DR Region where compute is only paid for during tests or a real failover. Once data volumes are confirmed (question Q1, Appendix A), the AWS Pricing Calculator should be used to produce a monthly estimate.

6. Migration and Operational Considerations

6.1 Migration Strategy

AWS describes seven migration strategies, the "7 Rs", and advises rehosting or replatforming first and modernising once workloads are running on AWS (Amazon Web Services, n.d.-e). Table 7 applies them to MedSys.

Table 7: Migration strategy per component

Component	Strategy	Method
File store (documents, reports, images)	Replatform to S3	AWS DataSync bulk copy, S3 File Gateway during transition
Portal servers	Replatform	Rebuild on EC2 from code; AWS Application Migration Service if a rebuild is not possible
Database	Replatform	AWS DMS full load then continuous change data capture (CDC) into RDS
Email platform	Repurchase	Replace with Amazon SES
Backup system	Replatform	Replace with AWS Backup, S3 versioning and Object Lock
Unused legacy systems	Retire	Export required records to S3, then decommission

6.2 Phased Plan

Table 8: Migration phases (indicative, based on A8)

Phase	Main activities	Clinic downtime
0. Plan and foundations (weeks 1–4)	Answer Appendix A questions; complete DPIA; build Control Tower landing zone, VPC, VPN, IAM Identity Center and central logging	None
1. Data first (weeks 5–10)	DataSync bulk copy of the file store to S3; S3 File Gateway so legacy apps write to S3; configure lifecycle, Object Lock and replication	None
2. Quick wins (weeks 8–12)	Move notifications to SES after domain verification; start AWS Backup plans	None
3. Build and test (weeks 10–18)	Build EC2 and RDS from code; DMS full load plus CDC; performance and failover tests with two pilot clinics	None
4. Cutover (one night)	Freeze changes; final DataSync and CDC catch-up; switch Route 53; smoke tests	Short planned overnight window (target under one hour)
5. Optimise and decommission (weeks 20–26)	Right-size, buy Savings Plans, run first DR test, retire on-premises kit after the rollback period	None

6.3 Minimising Downtime

- **Continuous synchronisation:** DMS CDC keeps RDS in step with the on-premises database, and DataSync incremental runs copy only changed files, so the final sync takes minutes rather than hours.
- **DNS preparation:** lowering the Route 53 time-to-live to 60 seconds a few days beforehand means clinics switch almost immediately.
- **Pilot clinics first:** two clinics use the new platform for a week before everyone moves.
- **Overnight window:** cutover follows MedSys's existing out-of-hours maintenance practice, and clinics are told in advance.

- **Rollback plan:** the old environment is kept read-only for an agreed period, and DNS can be reverted if acceptance tests fail.

6.4 How IT Operations Will Change

- **From hardware to services:** no server rooms, disk purchases or warranties. The team focuses on application, data and security configuration.
- **Infrastructure as code:** every change is made through reviewed, version-controlled templates, and rolling deployments allow many changes during the day without downtime.
- **Monitoring and response:** CloudWatch alarms and Security Hub and GuardDuty findings are routed to an on-call rota with written runbooks.
- **Policy-driven backup:** AWS Backup plans replace manual jobs, and a monthly restore test is recorded as evidence.
- **Access and contractors:** staff and contractors get named identities with MFA and time-limited elevated access, all logged in CloudTrail, and work to the same standards.
- **Cost management and skills:** a monthly cost review uses tags and budgets. Core staff train towards AWS Cloud Practitioner and Solutions Architect Associate, with an AWS partner supporting the first six months.

7. Benefits and Risks

7.1 Benefits

Table 9: How the proposal addresses current issues

Current issue	Benefit after migration
Portal slows at busy times	Auto Scaling and direct-to-S3 uploads keep response times steady as demand rises
Unplanned storage growth	S3 grows automatically; lifecycle rules cut the cost of older files
Manual, untested backups	Automated, locked and regularly tested backups held in two Regions
Single points of failure	Every tier runs across two AZs, with pilot-light recovery in Ireland
Security exposure	Encryption, MFA, WAF, threat detection, personal data discovery and full audit logs
Staff time spent on maintenance	Managed services and infrastructure as code free time for new clinic features

7.2 Risks

Table 10: Risk register (L = likelihood, I = impact; L/M/H)

Risk	L	I	Mitigation
Misconfiguration exposes patient data	M	H	Account-level Block Public Access, Control Tower guardrails, AWS Config rules, Macie, peer-reviewed code
Breach of UK GDPR or residency requirements	L	H	DPIA, London primary Region, documented DR transfer, customer-managed KMS keys, AWS data processing terms

Risk	L	I	Mitigation
Migration disrupts clinics or loses data	M	H	Phased plan, continuous sync, pilot clinics, rollback plan, backups taken before each phase
Cloud costs exceed budget	M	M	Budgets and alerts, Savings Plans, lifecycle rules, monthly cost review
Skills gap in a small team	H	M	Training, partner support, runbooks and managed services
Internet outage at head office or a clinic	M	M	Dual internet links at head office; the AWS side is unaffected and clinics already rely on internet access
Dependence on one provider	M	L	Open formats (PostgreSQL, standard files in S3) and portable infrastructure code
Regional AWS outage	L	H	Pilot-light DR in Ireland, tested twice a year

8. Conclusion and Recommendations

MedSys's current environment has carried it to more than 40 clinics, but fixed capacity, manual backups, knowledge spread across a split team, and exposure to ransomware make it a weak base for further growth. A cloud strategy on AWS addresses each of these, provided MedSys keeps ownership of configuration, access and compliance under the shared responsibility model.

Amazon S3 is well suited to sit at the centre of that strategy. It removes storage capacity planning, protects clinical files with eleven-nines durability, versioning and Object Lock, lowers the cost of long retention through lifecycle tiers, and opens a path to analytics, medical imaging and AI services in future. MedSys should:

1. hold the IT manager meeting, confirm assumptions A1–A8 and complete a DPIA;
2. set up a governed AWS landing zone in London with security guardrails before any data moves;
3. move files to S3 first, using DataSync and S3 File Gateway, and switch backups to AWS Backup with Object Lock;
4. replatform the portal to EC2 Auto Scaling and RDS Multi-AZ, using a DMS-based low-downtime cutover;
5. invest in staff training, review cost and security monthly, and test disaster recovery at least twice a year.

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Appendix A: Clarification Meeting Questions and Assumptions

Table A1: Questions for the IT manager

#	Question	Why it matters
Q1	How much data is stored today (documents, reports, images) and how fast is it growing?	Sizes S3 storage tiers, transfer time and cost
Q2	What are peak concurrent users and current response times on the portal?	Sets Auto Scaling limits and instance sizes
Q3	Which operating systems, database engine and versions run the portal? Who owns the code?	Decides rehost, replatform or refactor
Q4	How much downtime and data loss can the business accept (RTO and RPO)?	Drives Multi-AZ and disaster recovery design
Q5	How long must each record type be kept? Do any contracts require UK-only storage or the NHS Data Security and Protection Toolkit?	Sets lifecycle rules, Object Lock retention and Region choice
Q6	How are backups taken, where is the external copy, and when was a restore last tested?	Establishes the real recovery position today
Q7	Which email platform is used, how many messages are sent daily, and where are mailing lists held?	Sizes Amazon SES and plans the email cutover
Q8	What internet bandwidth do head office and clinics have?	Determines migration speed and hybrid links
Q9	What cloud skills do staff and contractors have, and how is contractor access controlled?	Shapes the training and access model

Table A2: Assumptions made pending confirmation

ID	Assumption
A1	About 15 TB of files today, growing 20–25% a year as clinics join. Files are read often in the first few months and rarely after a year.
A2	The portal is a conventional three-tier web application (web/application servers and a relational database such as PostgreSQL) that can run on Linux virtual machines without major code changes.
A3	Clinics use the portal mainly between 08:00 and 18:00 UK time; overnight load is low.
A4	Target recovery time (RTO) of 4 hours and recovery point (RPO) of 15 minutes for the portal; uploaded clinical documents must never be lost.
A5	Records are kept for at least eight years after last treatment, in line with typical health-record retention schedules (NHS England Digital, 2023).
A6	Primary data must stay in the UK. A disaster-recovery copy in Ireland (EU) is acceptable because the UK treats the EEA as adequate, subject to MedSys's DPIA.
A7	Head office has an internet connection of at least 500 Mbps that can be used for migration outside peak hours.
A8	MedSys has no existing AWS accounts, and the migration budget allows around six months of phased work.